

Glass Transition Dynamics and Surface Layer Mobility in Unentangled Polystyrene Films

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Most polymers solidify into a glassy amorphous state, accompanied by a rapid increase in the viscosity when cooled below the glass transition temperature (T_g). There is an ongoing debate on whether the T_g changes with decreasing polymer film thickness and on the origin of the changes. We measured the viscosity of unentangled, short-chain polystyrene films on silicon at different temperatures and found that the transition temperature for the viscosity decreases with decreasing film thickness, consistent with the changes in the T_g of the films observed before. By applying the hydrodynamic equations to the films, the data can be explained by the presence of a highly mobile surface liquid layer, which follows an Arrhenius dynamic and is able to dominate the flow in the thinnest films studied.

For polymers, glass transition temperature, T_g , is an important technological parameter because it is the temperature at which most polymers freeze into a solidlike glassy state (1). Studies over the past 15 years show that the T_g of nanometer polymer films can vary substantially with the film thickness, h , when h is decreased below 50 nm (2–8). Although the T_g of polymer films has been found to both increase (6) and decrease (4–8) with decreasing film thickness, the latter has drawn vastly more attention because of the often bigger size of the effect, especially in freely standing polystyrene (PS) films (8). According to conventional views, the presence of a solid substrate can decrease the configurational space available for the polymers to perform translational motions. Frictional forces between the polymer and the substrate surface can also slow down the dynamics. On the other hand, relaxation of the constraints to molecular motions at the polymer free surface can enhance the mobility. The overall dynamics of the films, and hence the T_g , should be a result of the interplay between all these effects. The most cited model for the thickness dependence $T_g(h)$ of polymer films, namely the layer model (4, 6, 7, 9), portrays the films to consist of a highly mobile surface layer on top of a less mobile, largely bulklike inner layer. Although there have been increasing experiments supporting the existence of a surface mobile layer on polymer films, notably for PS (9–11), its physical properties, including how its thickness depends on temperature, T , and the polymer thickness, h , as well as the mechanism by which it alters the dynamics of the films at the glass transition, are still controversial.

Keddie *et al.* (7) hypothesized that the surface mobile layer exists below the bulk T_g ($T_{g,bulk}$), and when the temperature T is increased toward

$T_{g,bulk}$, its thickness diverges critically, following $\sim(1 - T/T_{g,bulk})^{-\nu}$, where ν is a constant between 0.56 and 1 (2–4, 6, 7, 12–14). As a result, thinner films can be filled with the surface mobile layer and melt at a lower temperature. Herminghaus *et al.* (12, 13) proposed that the T_g of the films is determined by the fastest surface capillary mode that can penetrate the whole film. As the film thickness decreases, the required wave vector, and hence the relaxation rate of the fastest mode, increases. So, thinner films require a lower temperature to melt. To fit the $T_g(h)$ data, however, the model still requires the existence of a critically thickening surface mobile layer. Mechanisms that are completely independent of a free surface have also been proposed. By using molecular dynamics simulations, Varnik *et al.* (15) showed that an enhancement in the glass transition dynamics could be produced by confining the polymers between two repulsive, impenetrable walls. Long *et al.* (16) put forward another model on the premise that liquids undergoing the glass transition contain simultaneous fast and slow domains. Upon cooling, the slow domains grow at the expense of the fast ones; the glass transition

occurs when the slow domains percolate through the film. Because the percolation threshold is bigger in two dimensions than in three dimensions, the T_g can be lower in thin films. Although the above views for the mechanism of T_g reduction are diversified, they can all provide a good description to the observed $T_g(h)$ with an appropriate choice of the model parameters. Therefore, measurements additional to $T_g(h)$ are necessary to distinguish the prevailing mechanism.

The T_g of polymer films is usually determined by the discontinuous jump in the thermal expansivity in a temperature scan (4–7). Only a handful of experiments have measured the viscosity (6), which provides a straightforward measure of the rapid slowing down at the glass transition. In this study, we measured the viscosity of unentangled, short-chain PS films coated on silicon by monitoring the evolution of the surface structure of the films upon annealing by atomic force microscopy (AFM) (17, 18). The PS used has a weight-average molecular weight of 2.4 kg/mol and a polydispersity index of 1.06 (19). Figure 1, A and B, shows the topographic image of an $h = 4$ nm film upon annealing at 328 K for 2 and 24 hours, respectively. Although the temperature used in obtaining these images is 9 K below the T_g of the bulk polymer ($T_{g,bulk} = 337 \pm 2$ K, as determined by multiple thermal expansivity measurements on an $h = 120$ nm film upon cooling), visible features can be seen on the film surface within 2 hours of annealing, and by 24 hours, well-defined holes are developed across the film. These should be contrasted with the corresponding images obtained from an $h = 79$ nm film (Fig. 1, C and D), where no morphological development is discernible with the same annealing conditions. To understand the difference, we examined the evolution in the power spectral density (PSD) of the $h = 4$ nm films upon annealing. Figure 1E shows a sequence of PSD (open circles) obtained from one annealed at 334 K. The solid lines represent the best fit of the data to equations (S1a and S1b) derived from a model assuming the surface dynamics to be

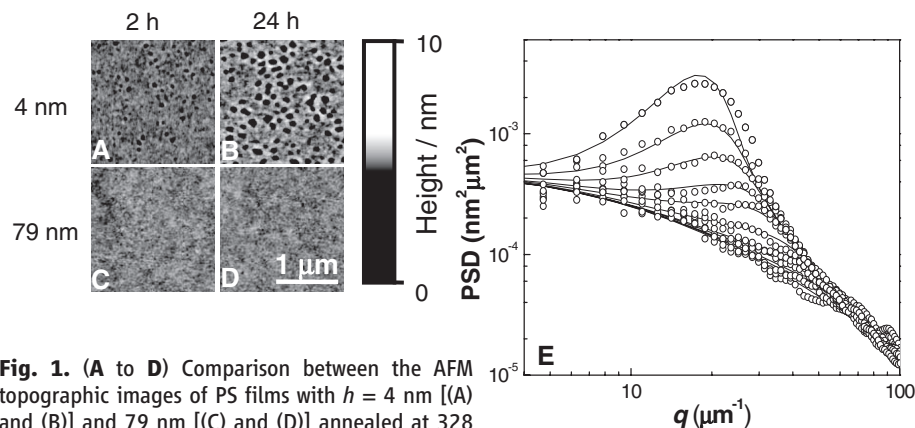


Fig. 1. (A to D) Comparison between the AFM topographic images of PS films with $h = 4$ nm [(A) and (B)] and 79 nm [(C) and (D)] annealed at 328 K and annealing times of 2 and 24 hours, respectively. Scale bar, 1 μ m. (E) Power spectral density of a 4-nm film upon annealing at 334 K for various times (from bottom to top): 0, 15, 30, 60, 90, 210, 600, 1080, 2100, 3840, and 7200 s (open circles). The solid lines denote the least-square fit of the data to eq. S1, as detailed in (19).

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governed by thermal equilibration of the surface capillary modes. The value of the viscosity of the film, η , deduced is only $\sim 10^{-5}$ times the bulk value. On the other hand, a low viscosity value is in keeping with the large disparity in morphological development noted between the two films.

Figure 2A depicts the measured η versus T of PS films with various thicknesses (symbols). The viscosity of the thickest ($h = 79$ nm) film displays an excellent agreement with the published value of the bulk polymer (20) (dashed line). In contrast, the viscosity of the thinner films is significantly reduced, especially at low temperatures. The temperature dependence of the viscosity of a glass-forming liquid is conventionally related to its T_g by the Vogel-Fulcher-Tammann (VFT) relation (1).

$$\eta(T) = \eta(\infty) \exp\left(\frac{B}{T - T_K}\right) \quad (1)$$

In Eq. 1, B is a constant and T_K is the Kauzmann temperature, that is, the theoretical temperature at which the configurational entropy of an ergodic, supercooled liquid vanishes (1). For our polymer in bulk, T_K is 288 K, or 49 K below $T_{g,bulk}$, and B is 1620 K (20). As an initial attempt to connect our result to the T_g , we fit the VFT relation to the data in Fig. 2A and display the result by solid lines. In obtaining the fits, we varied both T_K and $\eta(\infty)$ while keeping B equal to the bulk value. The fitted lines describe the data reasonably well. Shown in Fig. 2B are the fitted values of T_K for different films. As seen, they are almost constant, close to $T_{g,bulk} - 48$ K at large h , but decrease rapidly with decreasing h for $h < \sim 20$ nm. We find that the data of T_K can be fitted to the equation (solid line in Fig. 2B)

$$T_K(h) = \frac{T_g(\infty)}{(1 + h_0/h)} - \Delta T \quad (2)$$

where $T_g(\infty)$, ΔT , and h_0 are constant fitting parameters. The first term on the right side, which incorporates the dependence in h , coincides with the generic $T_g(h)$ dependence found for all supported (12, 13) and freely standing films with

$M_w \leq 378$ kg/mol (4). The fitted value of h_0 , 0.6 ± 0.1 nm when $T_g(\infty)$ is fixed to $T_{g,bulk}$, matches reasonably well with that obtained by modeling the experimental $T_g(h)$ data to $T_g(h) = T_g(\infty)/(1 + h_0/h)$ (12). The fitted value of ΔT (48 ± 3 K) also agrees well with the value known for the bulk polymer. Our result thus shows that the reduction in $T_K(h)$ by viscosity measurements agrees quantitatively with the reduction in $T_g(h)$ by thermal expansivity measurements. Notwithstanding the good agreement between the measurements and Eqs. 1 and 2, the ensuing implications are limited because the physical origins of Eqs. 1 and 2 are largely empirical and still controversial (1, 13).

Our main analysis begins when we replot the viscosity data as η/h^3 versus $1/T$ in Fig. 3A. We find that all the data deviating from the bulk η - T curve collapse onto the straight line corresponding to the following Arrhenius dependence (solid line in Fig. 3A)

$$\frac{3\eta}{h^3} = (165 \pm 7 \text{ Pa} \cdot \text{s} \cdot \text{m}^{-3}) \exp\left(\frac{(185 \pm 3) \text{ kJ/mol}}{RT}\right) \quad (3)$$

where R is the ideal gas constant. For ease of view, we have labeled all such data by filled symbols and the rest by open symbols. All the data for the $2.3 \text{ nm} \leq h \leq 9 \text{ nm}$ films collapse onto the Arrhenius line, whereas for the $h > 9 \text{ nm}$ films, that happens only at sufficiently low temperatures, with the onset temperature (denoted by the arrows in Fig. 3A) increasing with decreasing h . In Fig. 3B, we replot the data as η versus $1/T$ while preserving the same data-labeling scheme. One can see that the data that do not fall on Eq. 3 (open symbols) collapse into the bulk curve noted above. These observations suggest that the films can be separated into two regions, with one responsible for the collapse to the Arrhenius dependence seen in Fig. 3A and one for the collapse to the bulk curve seen in Fig. 3B.

We consider a bilayer film consisting of a homogeneous mobile layer at the free surface (with viscosity η_t and thickness h_t) and a bulklike inner layer (with viscosity η_b and thickness $h_b \equiv h - h_t$). In this experiment, the surface topography on the film produces inhomogeneities in

the Laplace and conjoining pressures and thereby pressure gradient, ∇P , parallel to the film (21). To find the flow pattern of the bilayer, we solve the Navier-Stokes equation for the system under a uniform pressure gradient, ∇P , parallel to the film, which is valid here as $qh \ll 1$. By assuming the no-slip boundary condition at both the bottom and intermediate interfaces and the interfacial tension between the two layers to be zero, we find the fluid velocity profile in the film, $v(z)$, to be given by:

$$v(z) = \frac{-\nabla P}{2\eta_b} [z^2 - 2(h_b + h_t)z] (z < h_b)$$

$$\text{and } v(z) = \frac{-\nabla P}{2\eta_t} \left[(z - h_b)^2 - 2h_t(z - h_b) - \frac{\eta_t}{\eta_b} h_b(2h_b + h_t) \right] (z > h_b) \quad (4)$$

A typical solution described by Eq. 4 is shown in the inset of Fig. 3B. The total (flow) mobility of the film, defined as $(\int_0^h v(z) dz)/(-\nabla P)$ (22), is given by

$$M_{\text{tot}} = \frac{h_b^3}{3\eta_b} + \frac{h_t^3}{3\eta_t} + \frac{h_t h_b (h_b + h_t)}{\eta_b} \quad (5)$$

The first two terms are simply the mobility of individual layers (denoted by M_b and M_t below) if the other layer were absent. The third is a coupling term, which under the experimental conditions is found to be always negligible. So our result shows that $M_{\text{tot}} \cong M_b + M_t$. In this experiment, what we measure is the relaxation rates of the capillary modes, which are directly related to M_{tot} (see eq. S1b); the viscosity, on the other hand, is a derived parameter, calculated by using $\eta = h^3/(3M_{\text{tot}})$ (21, 23). It follows that in a bilayer film, the measured viscosity is an effective one given by

$$\eta_{\text{eff}} \equiv \frac{h^3}{3M_{\text{tot}}} \cong \frac{h^3}{3(M_t + M_b)} \quad (6)$$

From Eq. 6, we see that the quantity being plotted in Fig. 3A is $\sim M_{\text{tot}}^{-1}$. The fact that the data for all the films with $h \leq \sim 9$ nm collapse onto the same line means that the M_{tot} of those films is the same, independent of h . Independence of M_{tot} on h means that the thinner films transport materials as effectively as the thicker films, contrary to the convention that they should be less effective. Our result can be explained if the mobility M_{tot} is dominated by the top layer so that $M_{\text{tot}} \cong M_t$. Any other explanation must entail the artificial relation $\eta_{\text{eff}} \sim h^3$ and thereby more contrived arguments.

We have used Eq. 6 to calculate the expected values of η_{eff} by using the experimental film thickness, values of M_t^{-1} given by Eq. 3, and $M_b = h^3/(3\eta_{\text{bulk}})$, where η_{bulk} is the viscosity of the bulk polymer (17, 20). The result, displayed by the solid lines in Fig. 3B, agrees remarkably well with the measurements without any adjustable parameters. This simple result shows that the two divergent trends (namely either Arrhenius or bulklike) found of the viscosity of the films are caused by the total mobility M_{tot} being almost

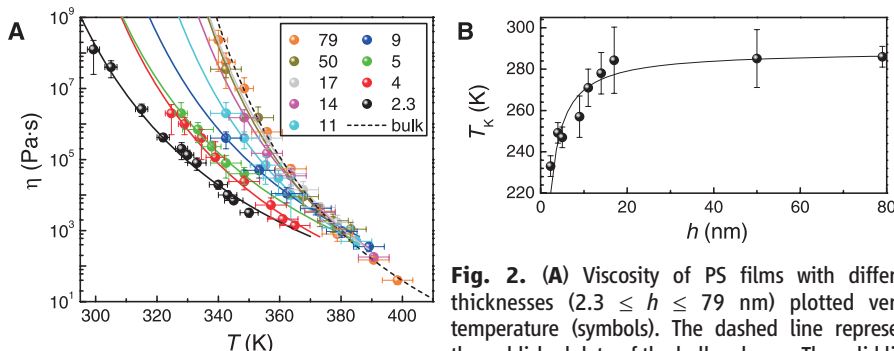


Fig. 2. (A) Viscosity of PS films with different thicknesses ($2.3 \leq h \leq 79$ nm) plotted versus temperature (symbols). The dashed line represents the published data of the bulk polymer. The solid lines are the least-square fits of the VFT relation to the data. (B) The fitted values of T_K obtained by fitting the data shown in Fig. 2A plotted as a function of the film thickness (solid circles). The solid line is the least-square fit to Eq. 2.

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always dominated by either M_t or M_b . Such a drastic crossover from the $M_{\text{tot}} \approx M_b$ to the $M_{\text{tot}} \approx M_t$ behavior is attributable to the vastly stronger temperature dependence of $M_b(T)$ (VFT) than $M_t(T)$ (Arrhenius). In the above analysis, we have assumed the mobile layer to be thin, with $h_t \approx 0$, whereby $M_b \approx h^3/(3\eta_{\text{bulk}})$. Adopting instead a small, finite h_t only slightly alters the calculated values of η_{eff} , mostly in the limited region corresponding to $T \approx 380$ K and $h \approx 11$ nm, where $M_b(T)$ and $M_t(T)$ have similar orders of magnitude. Although we have derived the model by assuming the films to be bilayers in which individual layers have uniform properties, the result (Eq. 6), involving only the mobilities of individual layers, can be completely independent of the layer structure. In particular, Eq. 6 embraces systems with a gradient layer structure, which has been suggested recently (9, 24).

In Eq. 5, the coupling term is not symmetric with respect to η_b and η_t . By exchanging the properties of the two layers, the model cannot fit well to the data at all. This can be traced to the fact that when the bulklike layer is riding on top of the mobile layer, the coupling term would become nonnegligible, whereby M_b cannot be approximated by $h^3/(3\eta_{\text{bulk}})$. This confirms the importance of the free surface in bringing about the reduction of the thin film T_g or enhancement in the glass transition dynamics, in disagreement with the percolation theory (16) and the possibility that the mobile layer may come from the polymer-substrate interface, as implied by simulation results (15). Our measurements are also incompatible with the models requiring the thickness of the surface mobile layer to diverge critically with temperature, as those discussed above. The fact that M_{tot} remains thickness independent among the $h < \sim 11$ nm films down to $h = 2.3$ nm (Fig. 3A, solid symbols) indicates that the surface mobile layer does not extend more than 2.3 nm from the free surface, because otherwise $M_{\text{tot}} (\approx M_t)$ there should decrease with

decreasing h as the surface mobile layer gets trimmed down. It can, however, be thinner if a third, dynamically dead layer (25) exists next to the substrate, for example. The coefficient, 165 ± 7 Pa·s·m⁻³ displayed in Eq. 3, is reliant on the total amount of material contributing to the Arrhenius transport and thus dictated by the effective surface layer thickness h_t . Constancy of its value, as demonstrated by the good fit of Eq. 3 to the data (Fig. 3A, solid symbols), implies that there is at most a weak dependence of h_t on T and h . This finding is in good accord with the result of Kawana *et al.* (25), who measured the thickness variation of the thermal expansion coefficient of PS supported by silicon and found that their data was consistent with a surface mobile layer with a constant thickness. We further note that the fluorescence-label experiment of (9) has found that the surface mobile layer extends no more than 33% into the film at $T = T_{g,\text{bulk}} - 5$ K, contrary to expectation if it had undergone critical thickening (7, 13). Nevertheless, critical thickening of a mobile layer at a water-surfactant-PS interface, differing from that studied here, is supported by a recent experiment (26). From our analysis, the surface mobile layer is able to dominate the flow in the $h < 9$ nm films at temperatures below ~ 385 K (Fig. 3A) because its viscosity at $T = 385$ K is at most 1/40 that of the bulklike inner layer, and the difference heightens rapidly with decreasing temperature. On the other hand, the thermal expansivity of the surface mobile layer was estimated to be only about 4 times that of the bulklike inner layer and does not change with temperature (25). Given the large discrepancies that can exist between different properties, the T_g of PS films obtained by different techniques can be different, which may explain some previous conflicting results (27).

It is remarkable that the Arrhenius dependence of $M_t(T)^{-1}$ is robust and persists from 306 to 400 K and over 7 orders of magnitude (Fig. 3A). Typically, Arrhenius temperature dependence is a

signature of local dynamics. Our finding suggests that the molecules close to the free surface, with reduced coordination, are free of cooperative couplings, contrary to those in the bulk. Moreover, this cooperativity-free behavior continues down to 306 K, or 31 K below $T_{g,\text{bulk}}$. The temperature dependence we found of the surface-layer mobility (Eq. 3) is consistent with those measured by lateral force microscopy (10) and surface nano-hole recovery (11). The activation energy we obtained, 185 kJ/mol, agrees within 20% with the values of 230 (10) and 150 kJ/mol (11) revealed in those experiments, although we caution that the nano-hole recovery time constant of (11) became T -independent below 303 K, outside the temperature range examined here. Although the surface dynamics discussed here do not necessarily have any direct counterpart in the bulk due to different constraints that may pertain at the free surface, we note that our activation energy roughly coincides with that of ~ 199 kJ/mol found in bulk PS below $T_{g,\text{bulk}}$ (28) and is of similar magnitude to that for the β relaxation (~ 140 kJ/mol), which has been attributed to dynamics confined to a very limited extent in the polymer chain (29).

We have shown that a surface mobile layer, exhibiting Arrhenius dynamics, with a thickness of less than 2.3 nm, is responsible for the reduction in the effective viscosity and, thereby, the T_g of unentangled, short-chain polystyrene-supported films. The thickness dependence of the T_g of the films found by viscosity measurement is in remarkable agreement with that of the T_g found by thermal expansivity measurement. Our result unfolds a mechanism by which the surface mobile layer can modify the overall dynamics of the films and does not require the surface layer thickness to be adjustable, thereby providing a more stringent test for the model. The other existing models, which either assume the surface mobile layer to be nonessential or to thicken critically with decreasing temperature, cannot explain our data.

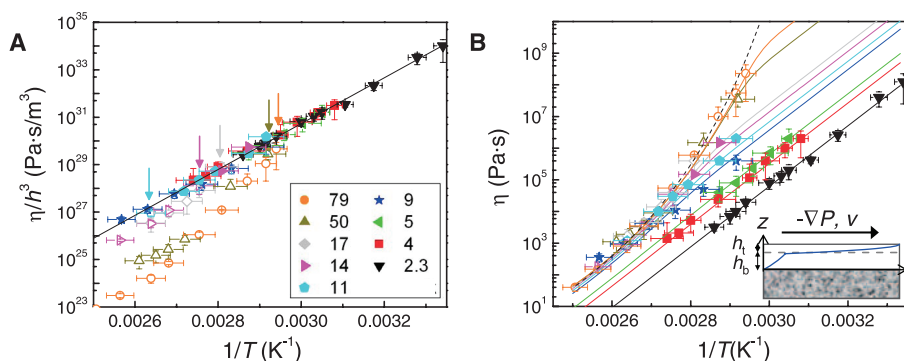


Fig. 3. (A) A plot of η/h^3 versus $1/T$ (open and solid symbols) for the same set of film thicknesses shown in Fig. 2A. The solid symbols label the data that collapse into the straight line shown by the solid line corresponding to Eq. 3. The open symbols label the rest of the data. The arrows indicate the temperatures at which the data of different thicknesses begin to depart from the solid line. (B) A plot of η versus $1/T$ (symbols). The same scheme used in Fig. 3A for the symbols applies here. The solid lines are generated from Eq. 6. (Inset) Schematic illustrating a simple two-layer model with homogeneous layer properties as discussed in the text. The blue curve represents the velocity profile for the fluid flow produced in the film by a uniform pressure gradient, ∇P , predicted by Eq. 4.

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Supporting Online Material

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Materials and Methods
Figs. S1 and S2
References

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The Palladium-Catalyzed Trifluoromethylation of Aryl Chlorides

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The trifluoromethyl group can dramatically influence the properties of organic molecules, thereby increasing their applicability as pharmaceuticals, agrochemicals, or building blocks for organic materials. Despite the importance of this substituent, no general method exists for its installment onto functionalized aromatic substrates. Current methods either require the use of harsh reaction conditions or suffer from a limited substrate scope. Here we report the palladium-catalyzed trifluoromethylation of aryl chlorides under mild conditions, allowing the transformation of a wide range of substrates, including heterocycles, in excellent yields. The process tolerates functional groups such as esters, amides, ethers, acetals, nitriles, and tertiary amines and, therefore, should be applicable to late-stage modifications of advanced intermediates. We have also prepared all the putative intermediates in the catalytic cycle and demonstrated their viability in the process.

The introduction of the strongly electron-withdrawing trifluoromethyl group into organic molecules can substantially alter their properties (such as lipophilicity, metabolic stability, and bioavailability) that affect the use of these molecules as pharmaceuticals and agrochemicals (1–3). Additionally, trifluoromethylated organic compounds find applications as materials such as liquid crystals (2). Despite the importance of this substituent, no general catalytic method exists for the introduction of the CF₃ group onto functionalized aromatic intermediates (4).

Structurally simple benzotrifluorides are accessible by radical chlorination of toluene derivatives and subsequent chlorine-fluorine exchange under harsh conditions (5). The replacement of an aromatic halide by a CF₃ group via copper-mediated coupling proceeds under milder reaction conditions but is mainly limited to aryl iodides (6–16). A catalytic version of this process was recently reported, but only aryl iodides with electron-withdrawing substituents and some heterocycles are good substrates (17).

A palladium-catalyzed trifluoromethylation of aryl halides (Fig. 1) has the potential to overcome these limitations: The use of a trifluoromethyl

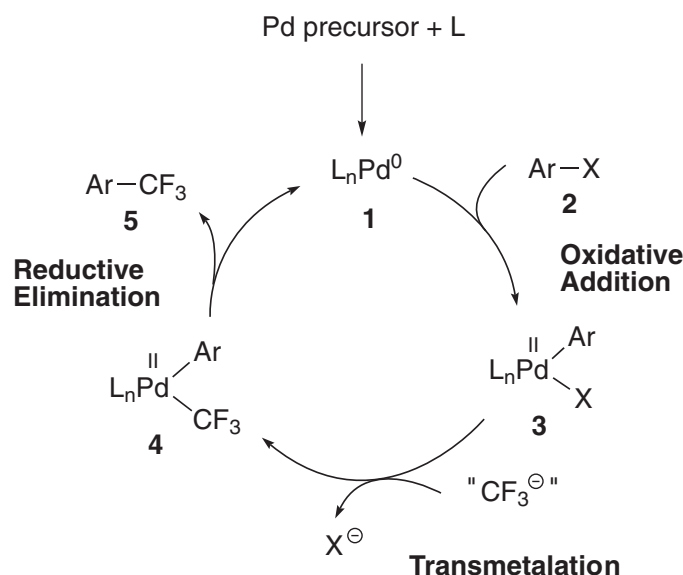
source as a transmetalating agent obviates the need for harsh reaction conditions that are required to replace individual substituents on benzylic carbon atoms with fluorine. Additionally, because many known ligands promote oxidative addition, even into unactivated aryl chlorides at low temperatures, a wide substrate scope is possible.

Mainly due to the high activation barrier for reductive elimination, the development of such a process has so far been unsuccessful. Several

complexes (4) with bidentate ligands yield either no (18, 19) or only trace amounts (20) of the benzotrifluoride products 5, even after prolonged heating at 130°C. The chelating biphosphine ligands 1,2-Bis(diphenylphosphino)ethane (dppe) and 1,3-Bis(diphenylphosphino)propane (dppp) promote the reductive elimination of 4, only at 145°C, to give PhCF₃ in 10 to 60% yield after 64 hours (19). On the other hand, the feasibility of fast Ar-CF₃ bond formation from a Pd(II) complex under mild conditions was demonstrated by Grushin and Marshall through quantitative conversion of the complex [XantphosPd(Ph)(CF₃)] to PhCF₃ upon heating to 80°C within 3 hours (21). However, the replacement of the Xantphos ligand in 3 with trifluoromethyl ions competes with transmetalation to 4, and consequently, no catalytic system with this system was reported (21, 22).

Complexes 4 are typically prepared from complexes 3, where X = Br or I, by treatment with TMSCF₃ (TMS, trimethylsilyl) and a fluoride source such as CsF, thereby using the formation of a silicon-fluorine bond as driving force (18–20). The challenge of using trifluoromethylsilanes in the presence of fluoride originates in the fluoride-initiated self-decomposition of R₃SiCF₃ to give R₃SiF and difluorocarbene (23). In a catalytic setting, where elevated temperatures are presumably required to promote reductive elimination from 4

Fig. 1. Generalized catalytic cycle for aryl trifluoromethylation (L = ligand; Ar = aryl; X = Cl, Br, I, triflate).



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Materials and Methods
Figs. S1 and S2
References

Glass Transition Dynamics and Surface Layer Mobility in Unentangled Polystyrene Films

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Materials and Methods

1. Sample Preparation

Polystyrene (PS), with a weight-average molecular weight, M_w , of 2.4 kg/mol and polydispersity index of 1.06, was purchased from Scientific Polymer Products (Ontario, NY). To make the polymer films, we dissolve the PS granules in toluene and filter the resultant solution through a PTFE filter with nominal pore size of 0.1 μm (Fisher Scientific Co.) before spin-coating it onto the substrates to form the films. The substrates, purchased from Siltronix (France), are single crystal (001) silicon wafers covered by a 102 ± 5 nm thick thermal oxide. Before use, the wafers are cut into $1.5 \text{ cm} \times 1.5 \text{ cm}$ slides and submerged in a piranha solution (a mixture of sulfuric acid and hydrogen peroxide in 7:3 volume ratio) at 130 °C for 10 min, followed by thorough rinsing in deionized water and drying by 99.99% nitrogen. Afterwards, the slides are further cleaned in oxygen plasma for 20 min whereupon ready for use. The thickness of the polymer film is controlled by either adjusting the concentration of the polymer solution or spinning speed, and monitored by ellipsometry at 5 different locations on the film. Deviation found in the measured thickness is less than 0.3 nm.

Most of the films were not annealed prior to measurement. We have found that pre-annealing has indiscernible effect on the measured viscosity (to be detailed below). During the viscosity measurement, the films are placed on a temperature-controlled hot plate inside a sealed chamber filled with 99.99% nitrogen at a slight positive pressure. The temperature is monitored by a thermocouple secured to a silicon wafer placed side by side the sample film. We find that the reading of the thermocouple can vary with its location on the hot plate within ± 5 cm from the center but not so much over time, with the amount of variation in the thermocouple reading increasing with increasing temperature. The error bars in Fig. 2A illustrate the maximum extent by which the thermocouple reading had varied.

2. Method to Measure the Viscosity

Previous studies showed that PS films on oxide coated silicon are unstable (S1, S2) and rupture into polymer beads upon heating (S2-S7). To determine the viscosity of the films, η , we follow the method detailed in Refs. (S8) and (S9). In brief, time-sequenced topographic images of the films are captured at various annealing times, t , by tapping-mode atomic force microscopy (AFM) as the films rupture. To convert the topographic data into power spectral density (PSD), $A^2_q(t)$, we first multiply the data by a Welch function before performing Fourier transformation and radial averaging, which results in the PSD, as those shown in Fig. 1E (S5). In this experiment, a local height fluctuation, $\delta h(\mathbf{r}, t)$, in the film produces a local excess pressure, $P(\mathbf{r}, t) = -\gamma \nabla^2 \delta h(\mathbf{r}, t) + G''(h) \delta h(\mathbf{r}, t)$ (S7, S10), where $\mathbf{r}$ is a position vector in the plane of the film, γ is the surface tension of the polymer and $G(h)$ is the van der Waals potential of the film. The excess pressure $P(\mathbf{r}, t)$ induces a two-dimensional polymer (flow) flux in

the film. In the long-wavelength limit, $qh \ll 1$, valid in this experiment, the planar flux is given by:

$$j(\mathbf{r}, t) = -M_{\text{tot}} \nabla P(\mathbf{r}, t) + \xi(\mathbf{r}, t),$$

where M_{tot} is the mobility of the film, defined by $[\int_0^h v(z) dz]/(-\nabla P)$ (S7, S10, S11), with $v(z)$ being the fluid velocity profile in the film instigated by ∇P , and $\xi(\mathbf{r}, t)$ denotes conserved noise arising from thermal activations. A linear stability analysis on the associated flow continuity equation predicts the following expression for the time-dependent PSD (S7, S10):

$$A_q^2(t) = A_q^2(0) \exp(2\Gamma_q t) + \left[\frac{k_B T}{G''(h) + \gamma q^2} \right] (1 - \exp(2\Gamma_q t)), \quad (\text{S1a})$$

$$\text{where} \quad \Gamma(q) = -M_{\text{tot}} [G''(h) q^2 + \gamma q^4], \quad (\text{S1b})$$

and k_B is the Boltzmann constant and T is absolute temperature. We have previously shown (S7-S9) that eqn. (S1a,b) describes the evolution of the surface of spin-coated polymer films very well. If the film is homogeneous, i.e., both its thickness h and viscosity η are uniform, and the no-slip boundary condition applies at the bottom interface, $M_{\text{tot}} = h^3/3\eta$. For generalization of M_{tot} to layered structures, please refer to the main text. To determine the viscosity of a film, we fit the $A_q^2(t)$ data of the film to eqn. (S1a,b) by using the viscosity parameter, η , as the only fitting parameter while keeping the other parameters fixed to the published and/or experimental values, after accounting for a low- q spectral background (S8). In performing the fits, we use only the data from the initial stage of roughening in the linear evolution regime well before the formation of any holes.

The first term on the right hand side of Equation (S1a) accounts for the conventional spinodal growth of the initial roughness of the film, $A_q^2(0)$. The second term accounts for thermal noise induced growth of the surface fluctuations. If the first term dominates, eqn. (S1a) reduces to the expression, $A_q^2(t) = A_q^2(0)\exp(2\Gamma(q)t)$, commonly used to describe the rupturing process of an unstable film (S4, S10, S12). Correspondingly, the PSD of the film, $A_q^2(t)$, would be peaked at the constant wavevector, $q_m \equiv \sqrt{-G''(h)/(2\gamma)}$, which maximizes $\Gamma(q)$ (n.b. $G''(h) < 0$ for unstable films). For the rupturing of polystyrene films on silicon, we have shown that the thermal noise induced fluctuations are significant (S9). From eqn. (S1a,b), the thermal noise induced fluctuations emerge systematically from high to low wavevectors. This explains the initial shift of the PSD peak seen in Fig. 1E and its subsequent convergence to $\sim 16 \mu\text{m}^{-1}$, which coincides with the estimated value of q_m .

Figure S1 shows that our method provides a sensitive measure for η . For a set of PSD's obtained from a $h = 4$ nm film annealed at 334 K, the model fit becomes visibly degraded when the value of η is adjusted by $\pm 12\%$ from the optimal value of 5.3×10^5 Pas. As a result, the value of the chi-square parameter increases by 9 to 26%. The error bars shown in Figs. 2A and 3 for the viscosity measurements represent the deviation from the optimal value of η that causes the quality of the fits to degrade visibly. It is remarkable that all the data, including those of the $h = 2.3$ nm films fit well to eqn. (S1a,b) even though 2.3 nm is comparable to the radius of gyration of the polymer (~ 1.3 nm). Since our model has assumed continuum hydrodynamics, this finding suggests that polymer layers with thickness down to ~ 2 nm can still be approximated by the continuum limit.

Previous studies had found that pre-annealing the films can have a significant influence on their dynamical behaviors (S13-S16). We have examined if pre-

annealing may affect our measurements by annealing several 4 nm films at 323 K, i.e., 32 K above the T_g of the films for 30 minutes in nitrogen before measuring their viscosity. The data obtained from one of the films measured at 334 K is shown in Fig. S2. The result shows that the annealed films give the same viscosity as the unannealed ones within experimental uncertainty. This is in keeping with our observation that the viscosities of the 79 nm films shown in Fig. 2A (which were not pre-annealed) are the same as the published data of the bulk polymer. This is also consistent with the fact that only a single value of viscosity is required to model the data obtained in all of our experiments, including those performed near or below the T_g of the bulk polymer, where the measurement time usually exceeds two hours (please see Figs. 1E and S2). These show that the majority of the structural relaxations, if any, in the films that may be caused by pre-annealing or annealing en route the experiment occur within an insignificant portion of the measurement time. Such a short structural relaxation time may arise from the polymer used here being made of short, unentangled chains, different from those used in the other studies (S13-S16).

3. Method to Determine T_K and its Error

The value of T_K is determined by least-square fitting the data of Fig. 2A to the Vogel-Fulcher-Tammann (VFT) relation (eqn. (1)):

$$\eta(T) = \eta(\infty) \exp\left(\frac{B}{T - T_K}\right),$$

where T_K and $\eta(\infty)$ are taken as the fitting parameters and B fixed to the bulk value of 1620 K (S17). The errors of T_K shown in Fig. 2A are determined by fixing $\eta(\infty)$ to the

best-fitted value while finding the maximum extent over which the value of T_K can be changed and still produce a model line that can pass through all the data points within their error brackets.

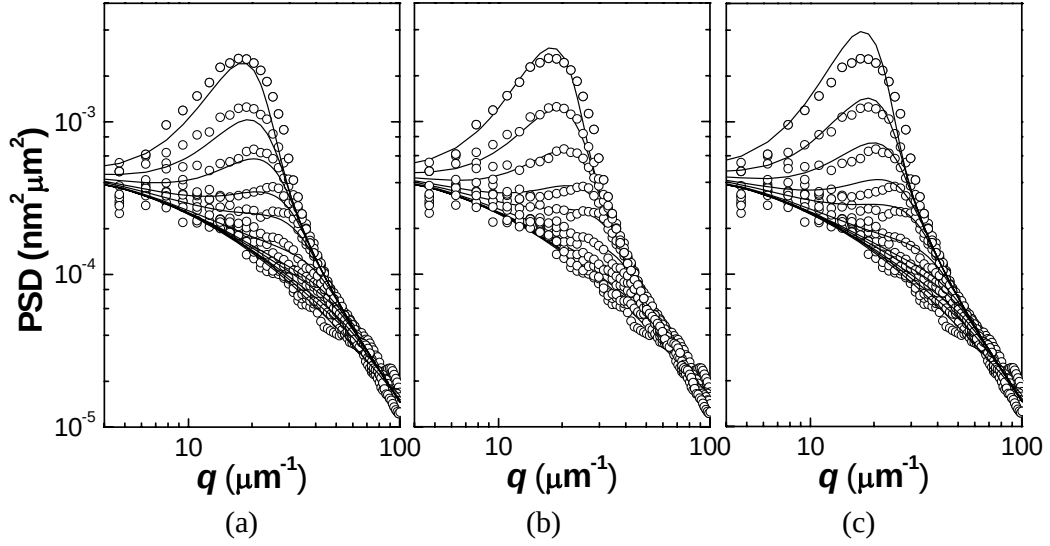


Fig. S1. Illustration of the sensitivity of the model fitting to variations in the value of η . The open circles represent the power spectral density, PSD, of a film with thickness, $h = 4$ nm after annealing at 334 K for times (from bottom to top) = 0, 15, 30, 60, 90, 210, 600, 1080, 2100, 3840 and 7200 s. The solid lines are calculations by using eqn. (S1a,b) with different values of η : (a) 6×10^5 Pas, (b) 5.3×10^5 Pas (optimal), and (c) 4.6×10^5 Pas; the respective values of chi-square are: 1.24×10^{-5} , 1.14×10^{-5} , and 1.44×10^{-5} . The quality of the fit worsens noticeably when η is adjusted by less than 15% from the optimal value.

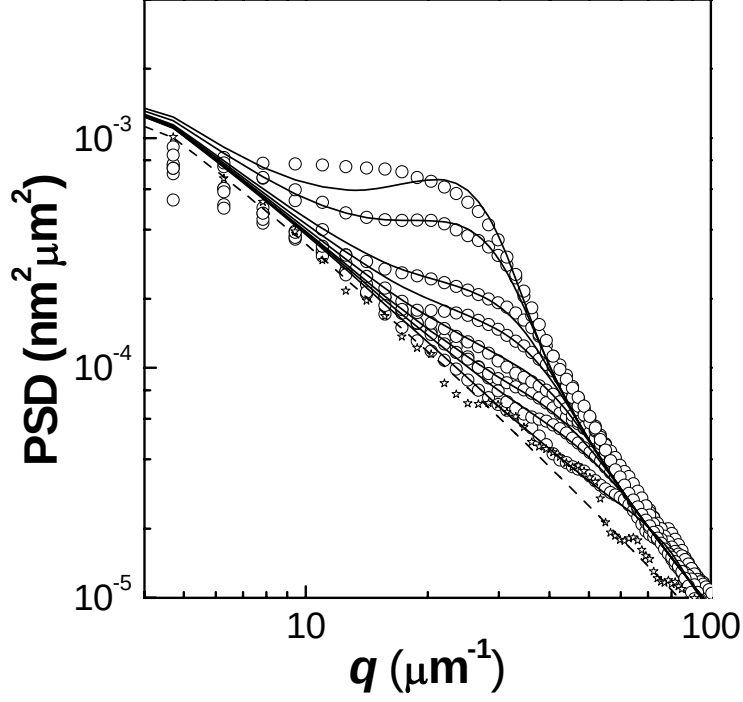


Fig. S2. Power spectral density obtained from a film with thickness, $h = 4$ nm, first pre-annealed at 323 K for 1800 s before subsequent time-sequenced measurement at 334 K. The stars are the data of the as-cast film. The dash line is to guide the eye. The open circles represent the data obtained upon heating at 334 K for times (from bottom to top) = 0, 20, 60, 120, 240, 480, 1080, 2100 s. The solid lines are the best fitted model lines. The fitted value of η here, being $(5 \pm 1) \times 10^5$ Pas agrees with the value of $(5.3 \pm 0.7) \times 10^5$ Pas obtained from the film not pre-annealed (Fig. S1 and 1E).

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